EXPERIMENT 2: DEPTH FIRST SEARCH (DFS)

AIM

To implement DFS traversal on a graph and explore all reachable nodes from a source.

PROCEDURE

1. Define the graph using an adjacency list.
2. Start traversal from the source node.
3. Recursively visit each unvisited neighbor until all reachable nodes are explored.

CODE (PYTHON)

```
def dfs(graph, start, visited=None):
    if visited is None:
        visited = set()
    visited.add(start)
    print(start, end=" ")

    for neighbor in graph[start]:
        if neighbor not in visited:
            dfs(graph, neighbor, visited)

# Sample graph
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

print("DFS traversal from node A:")
dfs(graph, 'A')
```

OUTPUT

```
DFS traversal from node A:
A B D E F C
```

RESULT

Thus, the DFS traversal was implemented successfully to explore all reachable nodes from the source node A.